Detecting Cafeteria Waste with Specialized YOLO11 Models

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Abstract

Automated cafeteria waste sorting involves two distinct tasks: identifying discarded objects and determining the disposal stream appropriate for each one. This study evaluates a prototype that combines four specialized YOLO11 detectors, covering seven cafeteria-item classes, with a rule-based policy that suggests composting, recycling, or manual review. The frozen models were evaluated on 186 validation photographs and a historically exposed cohort of 248 photographs containing 924 AI-assisted object annotations. On the latter cohort, the system achieved 87.78% mAP50 and 67.24% mAP50–95. At a uniform confidence threshold of 0.50, micro precision was 93.42% and recall was 78.35%. Selecting separate thresholds for each class on validation increased precision to 97.03% on validation and 97.22% on the historical cohort, while recall declined to 71.43% and 68.18%, respectively. Historical-cohort F1 fell from 85.23% to 80.15%. Only six historical images originated from the participating school, and the available records contain no independently verified disposal-bin annotations. The results therefore demonstrate how confidence selection changes recognition performance; they do not establish correct-bin accuracy or field generalization.

Keywords: object detection, YOLO11, cafeteria waste, selective prediction, recycling.

1 Introduction

At the end of a school lunch period, discarded food, drink cartons, utensils, and packaging may be mixed together on a tray or in a waste container. Computer vision can help identify these objects, but identifying an item and deciding where to discard it are not equivalent problems. A plastic bottle might be recognizable from its shape while its eligibility for recycling still depends on its contents, contamination, and local collection rules. The U.S. Environmental Protection Agency recommends checking local recycling guidance and emptying and cleaning recyclable containers [1]. A system that detects a bottle correctly may therefore still recommend the wrong bin.

Object detection is a useful starting point because it locates individual items in scenes containing several kinds of waste. Its predictions, however, involve uncertainty. A low-confidence prediction may be incorrect, while rejecting it may leave a real item unidentified. This distinction becomes important when detection results are used to support disposal decisions: a system can increase the proportion of correct predictions simply by withholding more of them, even as the number of correctly identified objects decreases.

This study examines a prototype consisting of four specialized YOLO11 detectors and a separate rule-based disposal policy. It addresses three related questions: how well the retained detectors recognize seven cafeteria-item categories on available development images; whether selecting confidence thresholds by class improves precision without retraining; and what additional evidence is required before object detections can be interpreted as reliable waste-sorting decisions. The analysis uses frozen checkpoints and recorded predictions, allowing changes in decision thresholds to be evaluated independently of changes in the detectors themselves. Because the nominal test photographs have prior development exposure, the findings are presented as development results rather than prospective field performance.

1.1 Related Work

Early approaches to waste recognition often classified the contents of an entire photograph. Thung and Yang's TrashNet project focused on individual waste items against comparatively simple backgrounds [2], while Single, Iranmanesh, and Raad introduced RealWaste using material photographed at a landfill [3]. These datasets illustrate how differences in image collection and object appearance affect the task being studied. Neither whole-image classification setting fully represents a cafeteria scene containing multiple overlapping items.

Localization datasets address a different aspect of waste recognition. Proença and Simões introduced TACO, which annotates litter instances within their surrounding environment [4]. Bashkirova and colleagues developed ZeroWaste to examine deformable and overlapping materials in industrial recycling imagery [5]. These studies are relevant to the crowded scenes encountered in waste collection, although recognizing an object or material remains different from verifying whether a particular school accepts it for recycling or composting.

The detector family used in this project originates from the YOLO framework introduced by Redmon and colleagues [6]. We use the YOLO11 implementation released by Ultralytics [8] and measure localization performance using the average-precision conventions associated with the COCO benchmark [7]. Average precision evaluates confidence-ranked detections over multiple operating points; it does not state how many correct and incorrect detections a system will retain after choosing a particular cutoff. For that reason, fixed-threshold precision and recall are also central to this study.

Selective classification provides a related framework for deciding when to withhold a prediction. Geifman and El-Yaniv examined confidence-based rejection [9] and subsequently introduced SelectiveNet, which jointly trains

prediction and rejection components [10]. In contrast, the present work does not train a rejection network. It applies different confidence thresholds to existing detectors and measures the resulting precision–recall tradeoff. Finally, research on distribution shift, including the WILDS benchmark [11], and on data leakage in scientific machine learning [12] highlights the importance of separating development decisions from independent evaluation. These concerns motivate our explicit audit of the historical image and annotation records.

2 Data

The working manifest contains 2,372 training views, 186 validation photographs, and 248 photographs in a historical test partition. The training count includes 2,052 original image entries and 320 additional augmented views. Twenty original photographs submitted by the participating school contribute 340 training views when their recorded augmentations are included. Augmented views present modified versions of existing photographs; they are not independent observations. Moreover, the four specialist checkpoints have different training histories, so the current manifest should not be described as a single common dataset used to train all four models from scratch.

The images were assembled from several public datasets and school submissions. The historical 248-image partition contains 73 images from a dataset labeled school_lunch, 66 from tetra_pak, 48 from trash_detection, 31 from plate, 20 from utensil, four from macworx, and six from the participating school's own collection. Thus, only 6/248 images (2.42%) were collected at that school; the separately named school_lunch source does not belong to the school's submission. The historical sample is consequently dominated by public-source imagery rather than representative observations of the intended cafeteria environment.

Table 1 summarizes the class distribution. The 248-image partition contains 924 annotated objects, but the distribution is highly uneven: food accounts for 433 objects, whereas hinged containers account for only four. The validation partition likewise contains just two hinged-container annotations. Strong average precision for such a sparsely represented class should therefore be interpreted cautiously.

The evaluation uses the retained, repaired annotation manifests rather than superseded label copies. A previous audit documented discrepancies in 109 older validation-label files, including ten files with bottle-label differences. Annotations were produced or reviewed with AI assistance, and comprehensive independent human verification is not documented. Ambiguous serving boundaries, visually similar packaging, and incomplete labels may consequently affect the measured results. No labels were modified for the present manuscript analysis.

Table 1: Object boxes by class (training views include augmentation; Hinge means hinged container).

Class	Train	Val.	Test
Food	2,501	343	433
Plate	722	110	123
Utensil	457	101	98
Hinge*	185	2	4
Carton	972	94	134
Wrapper	316	85	75
Bottle	293	42	57

3 Methods

The prototype combines four YOLO11 specialists, with each checkpoint responsible for a specified subset of the seven target classes (Table 2). One model detects food, hinged containers, and drink cartons; another detects plates and wrappers; the remaining two detect utensils and bottles separately. These divisions reflect the previously developed system. The present experiment does not establish that four specialists outperform a unified seven-class model, since no controlled comparison of those architectures was performed. Checkpoint hashes, class mappings, and inference settings were recorded in the research freeze at commit defcc6156a4425daf1ae530757634cc541f3c267.

Table 2: Frozen specialist models. Input size is given in pixels; Hinge denotes hinged container.

Model (input)	Parameters	Classes
YOLO11n (640)	2,590,425	Food, hinge, carton
YOLO11n (640)	2,605,126	Plate, wrapper
YOLO11x (800)	56,877,241	Utensil
YOLO11s (640)	9,428,179	Bottle

Inference was performed with Ultralytics 8.4.175, PyTorch 2.14.1, and Python 3.12. Each checkpoint processed the same image independently at its recorded resolution. Test-time augmentation was disabled; each specialist applied non-maximum suppression with an IoU cutoff of 0.70 and a limit of 300 detections. Outputs were then mapped to the seven shared class names and concatenated. Because no additional suppression was performed across specialists, overlapping predictions from different categories could remain in the combined output. This detail matters when one physical item resembles more than one target class.

The prototype includes a separate rule-based disposal layer rather than a learned bin classifier. At the default display threshold of 0.25, detections below confidence 0.50 are referred for review. Above 0.50, food detections receive a compost suggestion, while bottle and drink-carton detections receive a recycling suggestion. Plates, utensils, wrappers, and hinged containers are always marked for review. No category produces an automatic landfill suggestion. These rules are experimental: the system does not verify an item's material, whether a carton is empty, or the school's specific collec-

tion requirements. A suggestion therefore describes the rule applied to a recognized object, not independently established disposal eligibility.

The recorded evaluation environment was an Apple M1 MacBook Air with 8 GiB of memory and MPS acceleration available. Neither inference latency nor energy use was measured. The retained checkpoint and configuration hashes were checked during the research audit so that the results below refer to a fixed system rather than a new training run.

4 Evaluation

Object detection was measured using intersection-over-union (IoU), defined as the area shared by a predicted bounding box and its annotation divided by the area of their union. We report mAP50, which averages precision across classes at IoU 0.50, and mAP50–95, which additionally averages across IoU thresholds from 0.50 through 0.95 [7]. These ranking-based metrics are distinct from the precision of predictions retained at one chosen confidence threshold.

For the fixed-threshold comparisons, predictions were considered in descending confidence order and matched one-to-one to unused annotations of the same class at IoU at least 0.50. Each successful match counted as a true positive (TP); unmatched predictions and unmatched annotations counted as false positives (FP) and false negatives (FN), respectively. Micro precision, recall, and F1 pooled those counts across all seven classes:

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2PR}{P + R}.$$

An additional evaluation concerns the disposal-suggestion layer. For food, bottle, and drink-carton detections, classwise matching counts whether an automatic suggestion corresponds to an annotation of the same object class. A more conservative geometry-first match instead assigns predictions to annotated boxes before checking class agreement, exposing competition between overlapping specialists. Both methods measure object-recognition proxies: neither can measure correct-bin sorting without verified disposal labels. We also distinguish precision among accepted predictions from coverage of all annotated objects, because missed detections generate neither suggestions nor review requests.

The 186-image validation partition was used to choose operating thresholds. The 248-image historical partition was scored afterward without retuning those thresholds, but its photographs had already influenced earlier model development. Accordingly, these observations are not an independent test of generalization [12]. Confidence filtering changes the set of accepted detections and thus fixed-threshold precision and recall; it does not improve the checkpoint weights or the underlying ranking-based mAP.

5 Threshold Selection

We first evaluated whether a single confidence threshold could achieve high precision while retaining a useful number of detected objects. Global thresholds from 0.50 to 0.99, in increments of 0.01, together with 0.995, were evaluated on the 186-image validation partition. The selection criteria required at least 97% micro precision, 60% recall, and 100 accepted detections. No uniform cutoff met all three. Among the global thresholds satisfying the recall and detection-count requirements, the highest precision was 96.04% at confidence 0.79, with recall of 68.73%.

We next allowed the cutoff to vary by class. For each of the seven categories, candidate thresholds were summarized by the true and false detections they retained on validation. A dynamic program combined these possibilities, choosing the threshold vector that maximized accepted true positives while satisfying the same aggregate constraints:

$$\begin{aligned} \max_{\tau} \quad & TP(\tau) \\ \text{subject to} \quad & P(\tau) \geq 0.97, \\ & R(\tau) \geq 0.60, \\ & N_{\text{pred}} \geq 100. \end{aligned}$$

The resulting values appear in Table 3. Bottle received the strictest cutoff, 0.91, while utensil and hinged-container thresholds remained at 0.50. This method optimizes the mix of accepted detections on the validation set; it does not calibrate confidence scores as probabilities of correctness or guarantee that those thresholds will transfer to other image sources.

Table 3: Validation-selected confidence cutoffs by class; Hinge denotes hinged container.

Class	Cutoff	Class	Cutoff
Food	0.79	Plate	0.83
Utensil	0.50	Hinge*	0.50
Carton	0.84	Wrapper	0.83
Bottle	0.91	—	—

The selected vector was subsequently applied without modification to the historical cohort. It was not used to replace the existing disposal-policy settings, and no additional cutoff selection was performed on those historical images. Because those images had already informed development, the resulting comparison remains descriptive rather than an independent estimate of field performance.

6 Results

The frozen system achieved 87.78% mAP50 and 67.24% mAP50–95 on the exposed 248-image historical cohort. Corresponding validation scores were 86.36% and 65.25%. Tables 4 and 5 break down the historical results by class, with fixed-threshold precision, recall, and F1 evaluated at confidence 0.50 and IoU 0.50.

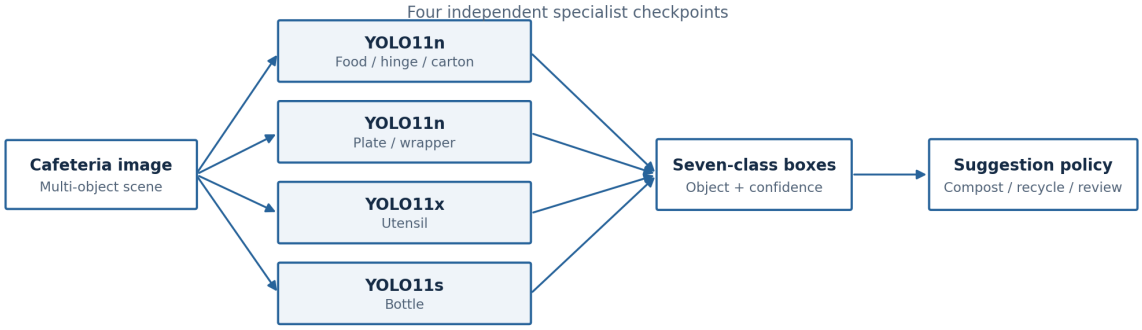


Figure 1: Processing pipeline: four YOLO11 specialists produce seven item classes, which are mapped to provisional compost, recycling, or manual-review suggestions.

Table 4: Detection precision, recall, and AP50 on 248 development images (%); Hinge denotes hinged container.

Class	Precision	Recall	AP50
Food	97.03%	75.52%	83.91%
Plate	87.29%	83.74%	87.04%
Utensil	90.91%	71.43%	86.06%
Hinge*	80.00%	100.00%	99.50%
Carton	99.23%	96.27%	97.55%
Wrapper	89.23%	77.33%	85.13%
Bottle	76.74%	57.89%	75.27%

Table 5: Detection counts, F1 and AP50–95 for the same cohort. TP/FP/FN are object counts; Hinge denotes hinged container.

Class	TP/FP/FN	F1	AP50–95
Food	327/10/106	84.94%	55.09%
Plate	103/15/20	85.48%	71.60%
Utensil	70/7/28	80.00%	61.26%
Hinge*	4/1/0	88.89%	91.17%
Carton	129/1/5	97.73%	74.28%
Wrapper	58/7/17	82.86%	57.64%
Bottle	33/10/24	66.00%	59.62%

Performance differed substantially across classes. Drink cartons produced 129 true positives, one false positive, and five false negatives, giving 99.23% precision and 96.27% recall. Bottles were more difficult: 33 true positives, ten false positives, and 24 false negatives yielded 76.74% precision and 57.89% recall. The food detector was precise when it returned a detection (97.03%) but missed 106 of 433 annotated food objects. Although hinged containers reached 99.50% AP50, that figure is based on only four annotated objects and cannot establish robust performance for the class. Across all seven categories, the uniform confidence-0.50 policy yielded 724 true positives and 51 false positives, corresponding to 93.42% micro precision, 78.35% recall, and 85.23% F1.

The difference between class-level and pooled measures is important. mAP gives each category equal weight in its macro-average, while micro precision and recall are dominated by classes with more annotated objects. Neither summary alone describes what happens when an individual category, such as bottle or hinged container, is underrepresented or difficult to recognize.

Table 6: Automatic-suggestion object metrics. Neither matching method establishes correct-bin accuracy.

Metric	Per class	Joint geometry
Matches / 510	489	483
Precision	95.88%	94.71%
Recall	78.37%	77.40%
F1	86.24%	85.19%

The disposal-suggestion layer uses a different operating policy from the classwise threshold experiment. Among its 510 automatic suggestions, conventional class-specific object matching identified 489 correct-class detections; a geometry-first assignment retained 483. The six disagreements involved five food detections and one bottle detection competing with wrapper annotations. This demonstrates that combining specialists without cross-class suppression can create ambiguous object assignments. It does not demonstrate six incorrect disposal-bin decisions, since no independently verified bin labels were available.

Table 7: Default sorting-policy outcomes. Denominators depend on detections versus labeled objects.

Outcome	Count (rate)
Compost	337/881 (38.25%)
Recycling	173/881 (19.64%)
All automatic	510/881 (57.89%)
Manual review	371/881 (42.11%)
Auto matched (IoU)	489/924 (52.92%)
Auto correct class	483/924 (52.27%)
Reviewed objects	286/924 (30.95%)
Missed objects	149/924 (16.13%)
Unmatched auto	21/510 (4.12%)
Unmatched review	85/371 (22.91%)

At the default display floor of 0.25, the pipeline produced 881 detections. Of these, 337 were provisionally assigned to compost, 173 to recycling, and 371 to manual review (Table 7). Thus, 57.89% of emitted detections received an automatic suggestion. This percentage is not coverage of actual waste items: the system cannot recommend or review objects it fails to detect. With all 924 annotated objects used as the denominator, only 483 objects (52.27%) were matched to automatic predictions of the correct visual class under geometry-first assignment. Even that figure is a recognition proxy, not evidence that those objects were sent to the correct disposal streams.

6.1 Precision–Recall Trade-off

The class-specific thresholds were selected using the validation partition and applied unchanged to the previously examined historical cohort. Table 8 compares this policy with a uniform cutoff of 0.50. The selected thresholds achieved the target of at least 97% validation micro precision, but their performance must be interpreted alongside the objects they excluded.

Table 8: Seven-class detection at IoU 0.50. P, R and F1 are percentages.

Cohort / cutoff	TP/FP/FN	P	R	F1
Val. 0.50	638/48/139	93.00	82.11	87.22
Val. classwise	555/17/222	97.03	71.43	82.28
Hist. 0.50	724/51/200	93.42	78.35	85.23
Hist. classwise	630/18/294	97.22	68.18	80.15

On validation, precision increased from 93.00% to 97.03%, while recall declined from 82.11% to 71.43%. The same pattern appeared on the historical cohort: precision rose from 93.42% to 97.22%, but recall fell from 78.35% to 68.18%. The historical false-positive count declined from 51 to 18, a reduction of 33, while true positives decreased from 724 to 630, a loss of 94 correct detections. Accepted predictions declined from 775 to 648 and F1 fell from 85.23% to 80.15%. Validation F1 likewise decreased from 87.22% to 82.28%.

These results show that the validation-selected policy is more conservative, not universally better. It produces fewer false detections among those accepted but leaves more annotated objects unrecognized. The optimization did not change the underlying checkpoint weights or mAP. Moreover, it has not replaced the default thresholds used to issue the disposal suggestions summarized in Table 7.

7 Data Independence

A file labeled as a test image is not necessarily independent of model development. A photograph may have been viewed during error analysis, included in earlier inference, duplicated elsewhere, or taken in a session already represented in training data. We therefore audited the retained image inventory and its recorded histories rather than assuming that the historical test folder represented a new sampling event.

The local inventory contained 102,806 readable file paths, 67,422 distinct inodes, and 14,379 byte-distinct images. Among 9,468 candidate image identities, 3,842 carried potential exposure or overlap flags, 5,482 had insufficient provenance to establish independence, and 144 original HEIC photographs could not be visually screened. No candidate was confirmed independent under the available records. This classification does not mean that every image was used by an optimizer; it means the audit could not establish the separation required for a new holdout.

Exact checks used file-level SHA-256 and decoded-pixel hashes, supplemented by rotation- and reflection-aware comparisons of 64-bit difference hashes within a Hamming distance of five. The screening pool included 4,751 images associated with training directories or manifests. These checks identify possible duplication and overlap, but cannot rule out shared capture sessions or related views merely because their hashes differ.

Table 9: Provenance of 9,468 candidate image identities. Categories do not imply confirmed training exposure.

Candidate status	Images
Exposure/group/duplicate flag	3,842
Unresolved provenance	5,482
HEIC decoding unavailable	144
Confirmed independent	0

The school-origin photographs require similar caution. The earlier and later submissions contained 123 and 21 photographs, respectively, and both shared recorded development or training capture groups. Four images from an October 29 submission were added to the training collection, so that capture session cannot serve as a prospective test. Historical phrases such as “sealed test” are therefore superseded by the later provenance records.

The annotation audit examined 2,398 JSON files in addition to CSV, XML, and YAML records. It found no confirmed collection of independently human-adjudicated, object-level disposal-bin labels. A prior “reviewed” flag does not, on its own, establish a human judgment about material composition or local bin eligibility. The audit also preserved 42 unreadable image aliases and five inaccessible or malformed JSON records as documented limitations rather than silently excluding them. Full diagnostics and protected-file hashes are recorded in results/research-readiness.json.

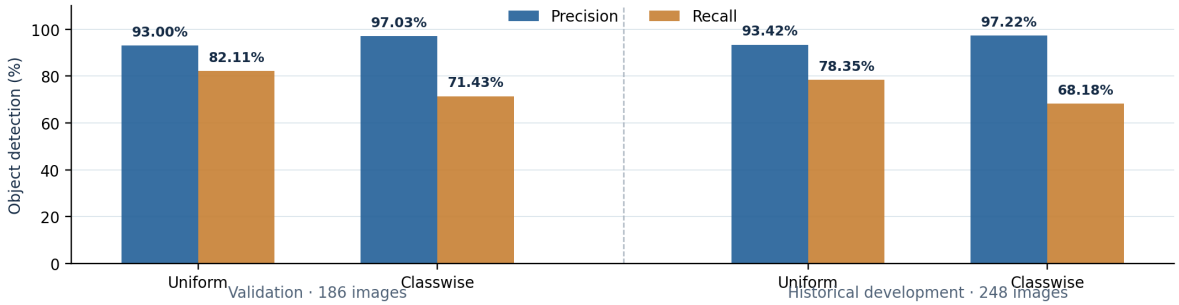


Figure 2: Precision and recall at uniform and class-specific thresholds, compared separately on validation and previously examined historical images.

Table 10: Independent sorting outcomes remain unmeasured; required evidence is shown.

Outcome	Required evidence
Bin precision/recall/F1	Independent images, human bin labels
Correct-bin accuracy	Verified local bin eligibility
Automatic object coverage	Complete independent annotations
Field review rate	Sealed school cohort

Table 11: Food and bottle error diagnostics at confidence 0.50. Categories are provisional.

Error category	Food	Bottle
Correct matches	327	33
FN: below threshold	53	19
FN: localization	48	1
FN: no overlap	5	4
FP: localization	2	0
FP: duplicate	1	0
FP: background/label	7	9
FP: class ambiguity	0	1

The missing measurements in Table 10 are not zero scores. They are undefined because the independent ground-truth information needed to calculate them was not collected or could not be verified. In particular, a high object-detection precision cannot be converted into sorting accuracy by assigning each detected class to its provisional rule-based bin.

8 Error Analysis

We examined the frozen predictions for food and bottle, two classes that contribute to automatic disposal suggestions. A missed object was assigned a diagnostic category according to the strongest same-class overlap in the saved outputs: a sufficiently localized detection below confidence 0.50, a candidate overlapping at IoU 0.10–0.49, or no qualifying same-class overlap. False positives were similarly grouped by localization, duplication, competing class, or lack of a corresponding annotation. These categories are algorithmic descriptions rather than verified causal explanations. In particular, an apparent background false positive could reflect an incomplete reference annotation.

Errors were concentrated in a few demanding photographs. One buffet scene accounted for 52 of 106 missed food objects (49.06%), while a single outdoor-trash scene accounted for 10 of 24 missed bottles (41.67%). The totals therefore reflect substantial within-image clustering rather than independent failures across dozens of different scenes. Both photographs were retained in the evaluation.

A targeted AI-assisted visual inspection considered 12 error cases from ten unique photographs. The examples included a sports-style bottle outside the intended disposable-container scope, a possible unannotated bottle, ambiguous cylindrical packaging, crowded food portions, and sealed food packaging recognized as food. These observations help identify plausible annotation and class-boundary problems, but cannot determine actual material, contents, or bin acceptance from the available previews. No annotations or disposal rules were modified.

A previously completed bottle-specialist trial also provided a negative experimental result. One additional epoch of classification-head training decreased validation AP50 from 82.47% to 77.95% and AP50–95 from 62.00% to 59.18%. The candidate failed its predefined validation criterion and was neither activated nor scored on the historical cohort. All main-paper results continue to use the frozen bottle checkpoint.

8.1 Additional Image Evaluation

The unchanged detectors were also applied to 44 source-screened public photographs: nine from bottle sources, six from carton sources, fifteen from utensil sources, twelve from hinged-container sources, and two from wrapper sources. At confidence 0.25, the system emitted 149 predictions—twelve compost suggestions, 49 recycling suggestions, and 88 review outcomes—while three photographs produced no displayed detection. These are output counts, not measurements against independently verified object or disposal labels.

A preliminary source-category check found at least one detection of the expected source class in 9/9 bottle photographs, 5/6 carton photographs, 12/15 utensil photographs, 4/12 hinged-container photographs, and

0/2 wrapper photographs. These proportions are not class recall: the source categories do not provide exhaustive instance-level ground truth, and several photographs contain multiple objects. Visual examination suggested repeated confusion between open foam containers and plates, missed wrappers, and overlapping predictions in crowded bottle scenes. In one example, a foil meal tray was classified as a drink carton with confidence 0.617 and consequently received a recycling suggestion; the photograph alone does not establish its correct local disposal stream.

A separate, provisional comparison used approximate AI-generated boxes for 41 of the 44 photographs, excluding three crowded bottle scenes. The frozen V1 model ensemble achieved 46.25% mAP50 and 21.16% mAP50–95, while a developmental hinged-container V3 candidate achieved 44.78% and 21.47%, respectively. Because the annotations were not independently human verified and the photographs had already been inspected during development, neither comparison constitutes a valid independent benchmark. The V3 candidate was not incorporated into the frozen system.

Taken together, these observations illustrate plausible failure modes and the limits of the available annotation process, not demonstrated field accuracy or a validated improvement under distribution shift. Case-level notes are preserved in the archived visual audit. The source images require rights verification before reproduction in a publication.

9 Limitations

The first limitation is evaluation independence. Although the model configuration and reported outputs are frozen, the historical cohort was consulted during earlier development, and the full lineage of every candidate photograph cannot be reconstructed. AI-assisted object annotations introduce further uncertainty, particularly in scenes where serving boundaries or similar forms of packaging are ambiguous. These factors weaken claims of generalization even when a benchmark's numerical precision appears high. The broader literature on distribution shift and leakage underscores why reused or mismatched evaluation data may overstate performance in a new setting [11, 12].

A second limitation is the gap between object recognition and sorting correctness. The policy does not establish whether bottles and cartons are empty, whether packaging contaminates food scraps, or whether the local waste contractor accepts a particular material. Nor does the system automatically recommend landfill disposal. Manual review is an abstention, not a successful sort, while undetected objects are omitted from the displayed review rate. No evidence was collected about contamination reduction, user compliance, inference latency, or energy consumption in a live school deployment. Public-dataset licensing and attribution must also be verified before any image redistribution.

An independent field study would require photographs collected in previously unused school sessions, with inclusion criteria and capture-group boundaries fixed before inference. Human annotators blinded to model predictions should identify each eligible object and document its actual disposal destination according to school-specific collection rules and, where necessary, material or contents evidence beyond the photograph. Disagreements and uncertain items should undergo recorded adjudication. The frozen checkpoints, thresholds, and decision rules could then be evaluated without further tuning, with class-specific detection metrics, correct-bin precision and recall, manual-review workload, and automatic coverage reported together. In addition to accuracy among accepted recommendations, the study should report the proportion of all eligible objects correctly assigned without review; otherwise, abstention could make a weak system appear artificially accurate. Uncertainty should be estimated across independent capture groups rather than treating objects from one image as independent observations.

10 Conclusion

This study evaluated a four-specialist YOLO11 system for recognizing seven categories of cafeteria waste. On 248 previously examined development photographs, the system achieved 87.78% mAP50 and 67.24% mAP50–95. Class-specific confidence thresholds increased object-detection precision to 97.03% on validation and 97.22% on the historical cohort, but lowered recall and F1. These results demonstrate the trade-off between accepting fewer false detections and missing more objects.

A central limitation is that the available images do not form an independent school-based test set, and verified bin labels are unavailable. The next step is to collect new cafeteria photographs and evaluate the unchanged system against independently reviewed object and disposal annotations. Only then can its effectiveness as a waste-sorting assistant be established.

11 Reproducibility

The project repository retains the inference code, class mappings, disposal rules, evaluation scripts, and summary results. The recorded configuration in configs/research-freeze.json identifies the four checkpoint hashes and fixed inference settings; results/sorting-verification.json contains the retained detection results, and results/research-readiness.json summarizes the provenance audit. Evaluation conventions and label requirements are described in docs/sorting-evaluation.md.

The complete image data, checkpoint binaries, and detailed local audit artifacts are held outside Git in the project's accompanying asset directory. The working data manifest and frozen repaired annotation files are likewise external. As a result, the public source code alone does not reproduce native inference. Historical source and license declarations are retained with the assets, but

attribution and redistribution permissions have not been independently audited; the school photographs are not presented as a released dataset.

The repository documents the commands for reproducing the file audit, running software tests, and evaluating newly collected annotations. These procedures do not retrain the frozen detectors or establish independent sorting accuracy; the latter requires a separately collected, human-verified evaluation cohort.

12 References

[1] U.S. Environmental Protection Agency. Frequent Questions on Recycling. Official recycling guidance. Accessed November 2026.

[2] G. Thung and M. Yang. Classification of Trash for Recyclability Status. Stanford CS229 Project Report, 2016. Report.

[3] S. Single, S. Iranmanesh, and R. Raad. RealWaste: A Novel Real-Life Data Set for Landfill Waste Classification Using Deep Learning. Information, vol. 14, no. 12, article 633, 2023. DOI: 10.3390/info14120633.

[4] P. F. Proença and P. Simões. TACO: Trash Annotations in Context for Litter Detection. arXiv:2003.06975, 2020. Preprint.

[5] D. Bashkirova et al. ZeroWaste Dataset: Towards Deformable Object Segmentation in Cluttered Scenes. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 21147–21157, 2022. CVPR paper.

[6] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi. You Only Look Once: Unified, Real-Time Object Detection. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 779–788, 2016. CVPR paper.

[7] T.-Y. Lin et al. Microsoft COCO: Common Objects in Context. European Conference on Computer Vision, pp. 740–755, 2014. DOI: 10.1007/978-3-319-10602-1_48.

[8] G. Jocher and J. Qiu. Ultralytics YOLO11. Software, 2024. Official model documentation and citation.

[9] Y. Geifman and R. El-Yaniv. Selective Classification for Deep Neural Networks. Advances in Neural Information Processing Systems, 2017. NeurIPS paper.

[10] Y. Geifman and R. El-Yaniv. SelectiveNet: A Deep Neural Network with an Integrated Reject Option. Proceedings of the 36th International Conference on Machine Learning, PMLR 97, pp. 2151–2159, 2019. ICML paper.

[11] P. W. Koh et al. WILDS: A Benchmark of in-the-Wild Distribution Shifts. Proceedings of the 38th International Conference on Machine Learning, PMLR 139, pp. 5637–5664, 2021. ICML paper.

[12] S. Kapoor and A. Narayanan. Leakage and the Reproducibility Crisis in Machine-Learning-Based Science. Patterns, vol. 4, no. 9, article 100804, 2023. DOI: 10.1016/j.patter.2023.100804.

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